The AI Energy-Demand Forecast Scorecard: what data-centre electricity forecasts measure and what can be checked

Purpose: describe a source-traced revision of the AI Energy-Demand Forecast Scorecard that extends the IEA 4E review of data-centre energy estimates to 2025 and 2026 headline forecasts.

NM AI Research. Version 0.4, 9 October 2026, revised 12 October 2026. Independent claim-support verdicts for every registered observation are recorded in SUPPORT_REVIEW.json; the candidate observations added in this version carry current supported receipts in CANDIDATE_SUPPORT_REVIEW.json.

This work was produced through an AI-assisted workflow directed and reviewed by the author.

Abstract

Forecasts of data-centre electricity demand from different publishers measure different quantities, draw different boundaries and date from different editions. Kamiya and Coroamă's critical review for the IEA 4E Technology Collaboration Programme catalogued over 100 publications by scope, time horizon and modelling approach and reported a much narrower range among the studies it rated higher in quality; Mytton and Ashtine traced the sources behind 258 published estimates. This revision of the AI Energy-Demand Forecast Scorecard extends that work to 2025 and 2026 headline forecasts from public bodies, research institutes, consultancies, banks and market analysts. Each registered figure is tied to a retained publication and an independent claim-support receipt, with dates, edition, target and base years, unit, quantity, scenario and coverage kept as separate fields. Candidate figures added in this version reach beyond the ledger's Global and US figures, to the Australian National Electricity Market, Europe and China as the IEA defines them, France and Denmark. A revision is computed only where one publisher states both figures for the same quantity and both carry supported receipts, and every declined comparison is listed. Few of the ledger's figures come from publications the 4E review catalogues. Spans of stated figures are narrower within like-for-like groups than across all figures for the same metric and year, but the narrowing comes from separating geographies, and with few observations per group the sample cannot show whether comparable forecasts converge. The paper makes no demand forecast, accuracy ranking or capacity-uplift estimate.

The question and why it matters

The scorecard asks two descriptive questions. What do selected 2025 and 2026 headline forecasts of data-centre electricity measure, and which comparisons between them can a reader support from the retained publications? And is the spread of their stated figures narrower within like-for-like groups, as the IEA 4E review found for the studies it rated higher in quality? It does not forecast electricity demand, rank forecast accuracy or estimate an uplift for announced capacity. Whether a difference between two published figures means anything depends on whether they measure the same quantity, on the same boundary, from the same base year; the sections below record what

each registered figure measures, which comparisons the inspected evidence supports, and how far apart the stated figures remain once unlike quantities are separated.

Earlier work

This version extends two reviews. Kamiya and Coroamă's critical review, prepared for the Efficient, Demand Flexible Networked Appliances platform of the IEA 4E Technology Collaboration Programme in March 2025 (P01 below), identified and reviewed over 100 articles, reports and statistics published since 2014. It catalogued each publication on affiliation, publication type, geographic scope, time horizon, modelling approach and analytical scope; recorded each study's values with its starting, ending and base years and one row per named scenario; removed cryptocurrency from the few assessments that included it; and, having found it challenging to define a rigid grading scheme, rated each study's quality by its two authors' independent judgement. Its recommendations ask modellers to define system boundaries and to avoid projections more than five years ahead, and ask journalists not to cherry-pick the most extreme scenario. Mytton and Ashtine (P02) traced 258 data-centre energy estimates from 46 publications through the 676 sources they used, argued that key sources must remain available so that calculations can be reproduced, and set out not to judge which estimate is more accurate.

Three publications from 2025 provide context. The guide by Koomey, Schmidt and Das for the Bipartisan Policy Center (P03) explains why regional load growth from data centres is hard to predict, including the overestimation that counting project proposals can cause; it states that the global estimates in its Figure C-1, which covers historical years, include only compute data centres and exclude network providers and cryptocurrency, compares projected growth rates with history, and judges projections of computing's electricity use more than a few years out to be no better than guesswork. London Economics International's report for the Southern Environmental Law Center (P04) evaluates projections of data-centre power demand growth by aggregating projections from US jurisdictions whose forecasts differ in issue date and periodicity; it interpolates years a forecast does not give at the cumulative average growth rate between two years, and extends a forecast whose outlook is too short by adding half of the change in demand between its last two years. The World Resources Institute (P05) asks why US forecasts differ so widely. Its explainer gives Lawrence Berkeley National Laboratory's range of 325 to 580 TWh as a 2030 figure, while the 4E review records the same range from LBNL's December 2024 report as a 2028 figure (P01, section 3.2.1), which shows why the target year is kept as its own field.

For Europe, ENTSO-E's May 2026 report on data centres and the power system (P06) gives estimated data-centre electricity demand by European country for 2024, 2030 and 2035, with a minimum for 2030 and a minimum and maximum for 2035, and states that demand in Europe is expected to grow by more than 50% between 2025 and 2030, citing the IEA, the European Data Centre Association and an Accenture analysis of estimates from multiple sources. This version instead registers each figure from the forecasting body's own retained publication, with its own claim-support receipt.

What this version adds is narrower than either review. It applies the 4E distinctions to a small set of 2025 and 2026 headline forecasts, including capacity and growth measures, whereas the 4E review's metric is annual energy; binds each registered figure to a retained publication and an independent claim-support receipt; refuses rather than harmonises comparisons across quantities, boundaries, base years and editions; and lists every comparison it declines. It is an update and extension, not a new method for comparing forecasts. The searches for closer work behind this statement, run on 9 October 2026 and, for European compilations, on 11 October 2026, were bounded and do not establish that none exists.

- P01. Kamiya, G. and Coroamă, V.C. (2025). Data Centre Energy Use: Critical Review of Models and Results. EDNA, IEA 4E TCP, March 2025. https://www.iea-4e.org/wp-content/uploads/2025/05/Data-Centre-Energy-Use-Critical-Review-of-Models-and-Results.pdf
- P02. Mytton, D. and Ashtine, M. (2022). Sources of data center energy estimates: A comprehensive review. Joule 6(9), 2032 to 2056. https://doi.org/10.1016/j.joule.2022.07.011
- P03. Koomey, J., Schmidt, Z. and Das, T. (2025). Electricity Demand Growth and Data Centers: A Guide for the Perplexed. Bipartisan Policy Center, 5 February 2025. https://bipartisanpolicy.org/report/electricity-demand-growth-and-data-centers/
- P04. London Economics International (2025). Uncertainty and upward bias are inherent in data center electricity demand projections. Prepared for the Southern Environmental Law Center, 7 July 2025. https://www.selc.org/wp-content/uploads/2025/07/LEI-Data-Center-Final-Report-07072025-2.pdf
- P05. Goldsmith, I. and Byrum, Z. (2025). Powering the US Data Center Boom: Why Forecasting Can Be So Tricky. World Resources Institute, 17 September 2025. https://www.wri.org/insights/us-data-centers-electricity-demand
- P06. ENTSO-E (2026). Data centres and the power system: expected trends, challenges and opportunities. European Network of Transmission System Operators for Electricity, May 2026. https://eepublicdownloads.blob.core.windows.net/public-cdn-container/clean-documents/Reports/2026/FINAL_ENTSO-E_Data_Centres_260430.pdf

Sources and vintages

The sample follows institutions in the existing scorecard and is not a census. The registered observations come from the IEA, LBNL, EPRI, McKinsey, Goldman Sachs, BloombergNEF, S&P Global with 451 Research, Deloitte and Gartner. The observation table carries the values, source locators, qualifications and review status. Historical editions remain distinct from refreshed observations. A later source in a different unit does not silently supersede an older annual-electricity estimate.

Deloitte's own publication supports attribution of its estimate to Deloitte even though its analysis draws on external material. This differs from relying on someone else's quotation of Deloitte. The public S&P report summary supports its stated estimate, while the complete model and the summary's cryptocurrency and enterprise boundaries remain unassessed.

Publication date, forecast edition, statement date, target year and historical base year are recorded separately. The held McKinsey capacity document is an August 2025 publication, correcting the earlier ledger's vintage. Deloitte's predictions edition names the following year but the underlying article was published in November 2024. The inspected Goldman primary summary dates its revision to March 2026, not April as recorded previously. The Goldman September 2026 episode was recorded in August 2026; its statement date and publication date are kept apart.

Where a report gives both a table and a rounded figure in its text, the ledger registers the tabulated value and records the text figure in the qualification. The IEA's 2026 report rounds its tabulated 2030 Base Case in the text, so the registered 2025 and 2026 editions are compared through their tables (O01 and O03, O02 and O18). A comparison between a tabulated value and a text figure is refused.

A currency sweep on 8 October 2026 looked for publications newer than each registered source. It added McKinsey's June, July and September 2026 statements (O21 to O23), Gartner's June 2026 release (O24 and O27), an S&P Global webinar report from June 2026 (O25) and Deloitte's April 2025 estimate (O26). Several of these pages refused automated retrieval; the author saved them in an ordinary browser session, and every quoted passage was matched against those copies. The following publisher-specific search outcomes are unverified from the bundled evidence; their search register is not included. Within the listed public-source searches, no newer qualifying forecast was located for the IEA, LBNL, EPRI or BloombergNEF. Goldman Sachs remained inconclusive because later secondary leads lacked located primaries. A newer McKinsey report lead was inaccessible and remains unverified; it is not registered here. Gartner's earlier ledger value and the BCG attribution remain unresolved. The report that a retained BCG article prints its range in gigawatt hours is also unverified here: U02 has no corresponding source-catalogue entry or support receipt. The legacy CSV preserves earlier rows without treating their historical verified labels as fresh support decisions.

The retained sources also state values the ledger does not register, such as the IEA's alternative 2030 cases, McKinsey's global annual-energy statement, Gartner's 2025 to 2027 consumption table and regional figures for PJM and Asia Pacific. They are listed with locators as known unregistered values and enter no calculation.

Candidate observations

For this version the registered publishers were searched on 9 October 2026 for publications dated after the 8 October sweep, and two editions from other publishers were added at the author's decision: TrendForce's press release of 16 September 2026 and AEMO's 2026 data-centre forecast with Oxford Economics Australia, published in August 2026. A figure is registered only where the publisher's own retained publication states a value with a target year, unit and geography that the ledger's fields can hold; a financial-year target is recorded under the source's own year label, with the financial-year basis stated. Other values in the same publications are listed as known unregistered values. At the author's request of 11 October 2026, figures from outside the United States were then added, European geographies first, under a rule declared before the new originals were retrieved: the registered publishers' European figures and those of European public bodies and transmission system operators, then China; a figure only from the forecasting body's own retained

publication dated 2025 or 2026, for 2030 where one is stated; the base, central, reference or median case, or else the case the publication presents as its headline; and each publisher's own definition of the geography. Three choices that the rule left open were settled after retrieval: a publisher's latest edition is registered and an earlier edition's figure is listed as a known unregistered value, because a candidate carries no supersession; where a publication states both annual electricity and capacity for 2030, the annual electricity value is registered, as the ledger does for the IEA; and a publication that names no central case but presents two retained trajectories is registered as the range across them, as the ledger records EPRI's scenario range. This added the IEA's tabulated 2026 Base Case values for Europe and China, RTE's figure for France and the Danish Energy Agency's figure for Denmark. The IEA's Europe is the European Union with other European countries, including the United Kingdom, Norway, Switzerland, Türkiye and Ukraine, so it is wider than the European Union and is not assumed to match another publisher's Europe. The full records, sources, known unregistered values and search records are in CANDIDATE_OBSERVATIONS.json. A candidate enters arithmetic, spans and declared comparisons, under the rules for a registered observation, only when CANDIDATE_SUPPORT_REVIEW.json holds a current supported receipt for it; one whose receipt is missing, stale or short of supported stays listed here with its status.

ID	Publisher, publication date	Stated figure	Geography and target	Quantity, scenario and boundary	Source
C01	TrendForce, 2026-09-16	490.7 GW	Global, 2030	capacity demand; scenario not stated; cryptocurrency not stated; total facility, IT plus overhead	N01
C02	AEMO with Oxford Economics Australia, 2026-08-25	15.5 TWh	Australia (National Electricity Market), FY30, a financial year as labelled by the source	consumption; Step Change; cryptocurrency not stated; facility boundary not fully stated	N03
C03	AEMO with Oxford Economics Australia, 2026-08-25	33.7 TWh	Australia (National Electricity Market), FY36, a financial year as labelled by the source	consumption; Step Change; cryptocurrency not stated; facility boundary not fully stated	N03
C04	IEA, 2026-04-16 (publication date on the IEA's report page)	114 TWh	Europe (IEA region), 2030	consumption; Base Case; cryptocurrency not stated; total facility, IT plus overhead	N06
C05	IEA, 2026-04-16 (publication date on the IEA's report page)	277 TWh	China (IEA region, including Hong Kong), 2030	consumption; Base Case; cryptocurrency not stated; total facility, IT plus overhead	N06

ID	Publisher, publication date	Stated figure	Geography and target	Quantity, scenario and boundary	Source
C06	RTE (Réseau de transport d'électricité), not verified (the URL path and file name carry 2025-12-23 and the PDF's metadata modification date is 23 December 2025)	10 to 15 TWh	France, 2030	consumption; décarbonation lente to décarbonation rapide (the two trajectories RTE retained after public consultation); cryptocurrency not stated; facility boundary not fully stated	N08
C07	Danish Energy Agency (Energistyrelsen), 2025-11 (printed month)	6 TWh	Denmark, 2030	consumption; AF25 base course (grundforløb), a target-fulfilment scenario; cryptocurrency not stated; facility boundary not fully stated	N09

Candidates C01, C02, C03, C04, C05, C06 and C07 have a current supported claim-support receipt and enter arithmetic, spans and declared comparisons as registered observations do. TrendForce states its figure as power demand capacity in gigawatts and includes non-IT equipment such as cooling and power supply; it describes its demand line as fleet electricity use converted into capacity on a basis it calls consistent with IEA public data, without giving a number. The AEMO and Oxford Economics figures are annual consumption in the National Electricity Market under the Step Change scenario, which AEMO describes as its most likely; their targets are financial years labelled FY30 and FY36, recorded under those labels rather than converted to calendar years, and AEMO's own overview reports the same figures approximately. The IEA figures are tabulated totals that include facility overhead; the workbook retained as the IEA's 2025 data annex gives 113 TWh for Europe and 277 TWh for China. RTE's figures are the 2030 data labels of a chart, 10 TWh in its décarbonation lente trajectory and 15 TWh in décarbonation rapide, for data centres on dedicated sites rather than inside the premises of the organisation that uses them; RTE states no facility boundary. The Danish figure is the base course of AF25, which the Danish Energy Agency describes as a target-fulfilment scenario. The IEA report, RTE's chapter and the Danish note do not state whether their figures include cryptocurrency mining. No publication dated 8 or 9 October 2026 from a registered publisher that states a qualifying figure was found; leads outside the rules, including European figures from ENTSO-E and Ireland's system operators that the rules do not admit, are listed in the search records.

Method

RESULTS.json records arithmetic, its denominators and the comparisons refused. A calculation runs only on observations whose current record carries a supported verdict from an independent reviewer; the Verification note describes how each verdict is bound to the records it covers. The drafting model's own assessment is recorded beside each observation and never enables a

calculation. A declared comparison involving an observation short of full support is refused, and RESULTS.json records which comparisons ran.

The generated groups show recorded-value spans among supported observations sharing a geography, metric, unit, target year and base year. Finite extrema exclude lower-bound records, which are listed separately; observation and publisher counts describe all group members. An earlier vintage superseded by a later vintage of the same publisher series is listed separately rather than counted in the span, so a span does not mix a publisher's revision with differences between publishers.

Method descriptions are limited to wording located in the retained sources, including server-shipment modelling and project-pipeline scenarios. They are qualified descriptions, not measured quality scores. Unknown scope is retained as unknown. Same-publisher lineage cannot override an explicit change in scope, scenario, unit, target or baseline.

The comparison with the 4E review first establishes, from the review's tables of reviewed studies and its reference list, which of the ledger's publishers and editions the review catalogues, and then sets what the review records for each publisher beside what the ledger records. Values attributed to the review are transcribed as it records them; they are not re-verified against the publications it reviewed, and no value is harmonised, converted or compared as a revision across editions.

The spread comparison sets each like-for-like group beside two wider sets for the same metric, unit and target year: the same geography with any edition or base year, and any geography. Target years match as each source labels them, so in the any-geography set a financial-year target counts under its labelled year. All three sets use only observations with a current supported receipt; registered observations short of one are excluded and listed in the table, and candidates without one appear only in the candidates section. Each span reports the lowest and highest stated bounds, lists open lower bounds separately and leaves scenarios, shapes and boundaries as stated.

Findings

A common unit and target year are not sufficient

The refreshed evidence does not support treating a common unit and target year as sufficient comparability. Annual electricity, installed capacity, capacity demand and growth relative to a base year are distinct quantities. Cryptocurrency coverage, facility boundaries, scenario construction, publication vintage and the precision of the stated figure can differ even within a nominally similar group. The ledger retains those distinctions instead of converting them through unstated utilisation or baseline assumptions.

Growth from a base year: Goldman Sachs

Goldman Sachs states its data-centre power demand outlook as growth from a base year and quantifies the underlying demand in gigawatts, so the ledger does not label the series as annual electricity. Its February 2025 statement is phrased as an upper end; the ledger records it as an upper bound with no central point, and it enters no calculation. The March 2026 summary names its

immediately prior outlook value, and the September 2026 transcript states the March figure on a 2025 base. The ledger registers both as source-stated vintages (O19 and O20). Identifying O20 as a restatement of O11 on a different base is an inference. The transcript may differ from the audio, which was not checked. Each declared revision pairs figures on one base: the March revision against its named prior on the 2023 base (which the package refuses because both observations are short of full support), and the September figure against the March figure on the 2025 base. Comparing percentages across the two bases would assign a direction to quantities with different denominators, and the executable comparison refuses it. An implied change in the forecast level holds the underlying baseline estimate fixed and is conditional on that assumption.

Installed capacity: BloombergNEF

BloombergNEF's later article names its December 2025 forecast as the comparator for the 2030 installed-capacity figure. For the 2035 figure the comparator is implied by date and the stated revision, and the December 2025 article words its figure as power demand. The current receipt for the December 2025 observation (O13) is partly supported because its installed-capacity classification is inferred. RESULTS.json therefore refuses the O13-to-O14 comparison (R03). A stated-point revision would require current supported receipts for both observations and would remain conditional on the inferred comparator. The records retain both the forecast-release date and the later article's publication date. Any such comparison would concern forecast change, not realised capacity growth.

The figures added by the currency sweep

Six of the seven figures added by the currency sweep (O21 to O27, a small descriptive group) are recorded in gigawatts: McKinsey's global capacity demand and US IT demand, its global capacity for 2035, Gartner's power demand, S&P Global's US capacity and Deloitte's electricity demand, assigned to the US by inference; the seventh is Gartner's 2030 consumption in TWh. Capacity demand, IT demand and capacity of unstated type are recorded as distinct quantities and are not converted to annual energy. McKinsey's June 2026 page repeats the rounded 2030 global figure of its August 2025 article (O09 and O21) while citing a December 2025 article that is not retained; without established lineage the ledger records both as separate observations, with no supersession. Gartner states its 2030 consumption only as over 1,200 TWh; the ledger records that as an open lower bound (O27), which enters no comparison and is listed separately in value spans rather than treated as a maximum.

A restated historical estimate: LBNL

LBNL's restated historical electricity estimate is kept outside the forward-forecast sample. Its later reference point lying within an earlier scenario range is an interval-containment observation. It does not establish an unchanged central forecast and is not automatically coded flat. Older numeric endpoints without fresh support remain in the preserved legacy ledger.

Disclosed input links among providers

Several registered providers disclose input links. The IEA's data-centre model relies on a bottom-up approach developed by LBNL; Goldman Sachs states that it uses a 451 Research data set; Deloitte's November 2024 estimate cites Goldman Sachs and EPRI among its sources, which predates their later registered observations, and its April 2025 estimate names LBNL's 2024 report among its inputs. Agreement among linked providers therefore cannot automatically be counted as independent confirmation; the exact shared inputs and vintages are not established.

Coverage in the 4E catalogue

The overlap is small. Of the ledger's 27 registered observations, 1 (O08) comes from a publication that the 4E review catalogues, 4 (O04, O05, O06 and O07) come from later editions of report series whose earlier editions it catalogues, and 22 come from publications it does not catalogue, although it catalogues other publications from all 9 registered publishers.

Publisher	4E entries and relation to the ledger	What the 4E review records	What the ledger records
IEA	IEA (2014) to IEA (2025); the latest are Electricity 2024, World Energy Outlook 2024 and Electricity 2025; other publications of the publisher only: O01, O02, O03, O18	Global, IEA (2024b) with Spencer and Singh (2024): 240 to 340 TWh in 2022 and a 2030 projection of 515 TWh (range 405 to 685 TWh). Scenario: range given; scenario names not given in the table. Base year: not given in P01's tables. Cryptocurrency: excluding cryptocurrency, as P01 records it. Quality: not assessed (lack of methodological details). (P01 Table 2.1, p. 5; Table 3.1, p. 16; references, p. 55)	O01: 946 TWh, Global, 2030, Base Case; cryptocurrency not stated; O03: 945 TWh, Global, 2030, Base Case; cryptocurrency not stated
LBNL	Shehabi et al. (2016, 2018, 2024); the latest is the 2024 United States Data Center Energy Usage Report; earlier edition of the same series: O04, O05, O06	United States, Shehabi et al. (2024): 178 TWh in 2023 and 325 to 580 TWh in 2028. Scenario: range given; scenario names not given in the table. Base year: not given in P01's tables. Cryptocurrency: excluding cryptocurrency, as P01 states for this report. Quality: high. (P01 Table 2.1, p. 5; section 3.2.1, p. 20; Table 3.2, p. 23; references, p. 60)	O04: 464 TWh, US, 2028, Reference Case; cryptocurrency excluded; O05: 521 to 843 (reference 649) TWh, US, 2030, Reference Case with Compounded Uncertainty bounds; cryptocurrency excluded; O06: 192 TWh, US, 2024, historical estimate; cryptocurrency excluded

Publisher	4E entries and relation to the ledger	What the 4E review records	What the ledger records
EPRI	EPRI (2024), Powering Intelligence; earlier edition of the same series: O07	United States: a 2030 projection of 200 to 405 TWh. Scenario: range from assumed growth rates; scenario names not given in the table. Base year: not given in P01's tables. Cryptocurrency: not stated for this entry; P01's US figure note says its US values exclude cryptocurrencies. Quality: low-medium. (P01 Table 2.1, p. 5; Figure 3.3 note, p. 21; Table 3.2, p. 22; references, p. 52)	O07: 380 to 790 TWh, US, 2030, EPRI scenario range; cryptocurrency included
McKinsey	McKinsey (2023), (2024a) and (2024b); McKinsey (2024a) carries the URL of ledger source S05; same publication: O08; other publications of the publisher only: O09, O21, O22, O23	United States, McKinsey (2024a): 147 TWh in 2023, 178 TWh in 2024 (25 GW capacity) and a 2030 projection of 606 TWh (80 GW). Scenario: not given in the table. Base year: not given in P01's tables. Cryptocurrency: not stated for this entry; P01's US figure note says its US values exclude cryptocurrencies. Quality: not assessed (no methodology disclosed). (P01 Table 2.1, p. 5; section 3.2.1, p. 20; Table 3.2, p. 23; references, p. 58)	O08: 606 TWh, US, 2030, medium scenario; cryptocurrency excluded by inference
Goldman Sachs	Goldman Sachs (2024), AI, data centers and the coming US power demand surge; other publications of the publisher only: O10, O11, O12, O19, O20	Global: a 2030 projection of 740 to 1,400 TWh; United States: 397 TWh in 2030. Scenario: Base Case, Bear and Bull (Figure 3.1 legend). Base year: not given as such; P01's method summary says the global projection builds on the high range of an IEA estimate for 2022. Cryptocurrency: not stated for these entries. Quality: global: medium; United States: not assessed. (P01 Table 2.1, p. 5; Figure 3.1, p. 13; Table 3.1, p. 15; Table 3.2, p. 22; references, p. 53)	O10: up to 165 per cent growth versus 2023, Global, 2030, scenario not stated; cryptocurrency not stated; O12: 170 per cent growth versus 2025, Global, 2030, scenario not stated; cryptocurrency not stated

Publisher	4E entries and relation to the ledger	What the 4E review records	What the ledger records
BloombergNEF	BloombergNEF, Statkraft and Eaton (2021), Data Centers and Decarbonization; other publications of the publisher only: O13, O14, O15	Germany, Ireland, the Netherlands, Norway and the United Kingdom: 26 TWh in 2021 for colocation and hyperscale data centres. Scenario: a 2021 estimate; no scenario given. Base year: not applicable (a 2021 estimate). Cryptocurrency: not stated for this entry. Quality: medium-high. (P01 Table 2.1, p. 5; Table 3.3, p. 25; references, p. 49)	O14: 194 GW, US, 2035, base case; cryptocurrency not stated; O15: 118 GW, US, 2030, base case; cryptocurrency not stated
S&P Global with 451 Research	S&P Global (2024), S&P Global Commodity Insights (2024) and S&P Global Market Intelligence (2024a), the 2024 US Datacenters and Energy Report; other publications of the publisher only: O16, O25	United States: Commodity Insights, 355 to 385 TWh in 2030; Market Intelligence, over 280 TWh in 2024 and 530 TWh in 2028; S&P Global, 340 to 420 TWh in 2030. Scenario: not given in the table. Base year: not given in P01's tables. Cryptocurrency: not stated for these entries; P01's US figure note says its US values exclude cryptocurrencies. Quality: Commodity Insights: medium; the other two: not assessed. (P01 Table 2.1, p. 5; Table 3.2, p. 24; references, p. 61)	O16: 728 TWh, US, 2030, base forecast; cryptocurrency not stated; O25: 151.734 GW, US, 2030, scenario not stated; cryptocurrency not stated
Deloitte	Deloitte Global (2024), Powering AI; other publications of the publisher only: O17, O26	Global: 380 TWh in 2023 and a 2030 range of 700 to 970 TWh. Scenario: Baseline and High adoption (Figure 3.1 legend). Base year: not given in P01's tables. Cryptocurrency: not stated for this entry. Quality: medium-high; P01 names it as the exception among consultancy and investment-bank studies. (P01 Table 2.1, p. 5; Figure 3.1, p. 13; Table 3.1, p. 14; section 5.2.2, p. 39; references, p. 51)	O17: 1,065 TWh, Global, 2030, conditional headline estimate; cryptocurrency not stated
Gartner	Gartner (2024a) and (2024b), among AI energy studies; other publications of the publisher only: O24, O27	AI energy use: 74 TWh in 2022, 195 TWh in 2023 and 500 TWh in 2027. Scenario: not given. Base year: not given in P01's tables. Cryptocurrency: not stated for this entry. Quality: not assessed (lack of methodological details). (P01 Table 2.1, p. 5; Table 3.4, p. 31; references, p. 52)	O24: 290 GW, Global, 2030, scenario not stated; cryptocurrency not stated; O27: over 1,200 TWh, Global, 2030, scenario not stated; cryptocurrency not stated

The differences are reported rather than harmonised:

- IEA: Different editions. P01's latest global IEA projection comes from the World Energy Outlook 2024; the ledger registers the tabulated Base Case values of the 2025 Energy and AI annex and the 2026 report, which P01 does not list. P01 records its IEA values as excluding cryptocurrency; the ledger records cryptocurrency coverage for its IEA rows as not stated. Not a revision comparison.
- LBNL: Earlier edition of the same report series: P01 covers the 2024 report and the ledger registers the 2025 Update. Both record cryptocurrency as excluded. The values come from different editions and are not compared as a revision.
- EPRI: Earlier edition of the same series: P01 covers the 2024 report and the ledger registers the 2026 brief. The ledger records the 2026 range as including cryptocurrency, while P01's US figure note says its US values exclude cryptocurrencies. Not compared as a revision.
- McKinsey: Same publication and the same 2030 value. P01 also records a 2030 capacity figure and 2023 and 2024 values, which the ledger does not register. The ledger records the medium scenario and a cryptocurrency exclusion inferred from the publication's footnote; P01's table states neither. McKinsey's four other registered figures come from later publications.
- Goldman Sachs: Different publications and quantities. P01 records annual energy in TWh from Goldman Sachs's 2024 report; the ledger registers later statements of power-demand growth relative to a base year, which it does not convert to energy.
- BloombergNEF: No overlap of edition, geography or quantity: P01 covers a 2021 European energy estimate, and the ledger registers BloombergNEF's 2025 and 2026 US installed-capacity forecasts.
- S&P Global with 451 Research: Earlier publications only. P01 covers S&P Global's 2024 reports; the ledger registers the 2026 outlook and a June 2026 webinar report. The 2024 report and the 2026 outlook have similar titles, but the retained records do not establish that they form one series, so they are not treated as editions of one series.
- Deloitte: Different Deloitte publications of late 2024. P01 records the 2030 range of Deloitte Global's Powering AI; the ledger registers the 2030 figure in Deloitte's 2025 predictions article, which lies outside that range. The retained records do not establish whether the two share a model.
- Gartner: Different publications and scope: P01's Gartner entries estimate AI energy use only, and the ledger registers Gartner's 2026 total data-centre figures.

The 4E catalogue and the ledger record different attributes of each figure:

4E catalogue parameter	4E values	Ledger fields	Relation
Affiliation	academia, government, industry, non-governmental organisation, mixed (P01 section 2.3, p. 6)	No field. The publisher is named in forecaster; its affiliation type is not recorded.	not recorded

4E catalogue parameter	4E values	Ledger fields	Relation
Publication type	peer-reviewed, preprint, report (P01 section 2.3, p. 6)	No field. The catalogue's source_type records whether a source is a primary document or table, not its publication type.	not recorded
Geographic scope	global, multi-country, country (P01 section 2.3, p. 6)	geography names the area covered: Global and US in the ledger; the candidates add the Australian National Electricity Market, Europe and China as the IEA defines them, France and Denmark.	recorded as a named area
Time horizon	historical, projection (P01 section 2.3, p. 6)	horizon records the target year; a historical estimate carries the metric historical_model_estimate and stays outside spans; forecast_vintage, source_publication and base_year record dates separately.	recorded at a finer grain
Methodological and modelling approach	bottom-up, aggregated totals, temporal proxy extrapolation, hybrid, other (P01 section 2.3, p. 6)	method_description records the method only in wording located in the source and assigns no class; model_reproduction records that no provider model was rebuilt.	described, not classified
Analytical scope	data centre types (enterprise, cloud, hyperscale) and specific technologies or topics (AI) (P01 section 2.3, p. 6)	coverage records the facility boundary and crypto_scope the cryptocurrency boundary; data-centre types appear in qualifications where the source states them.	recorded as boundaries
Recorded by P01 outside the six parameters	starting, ending and base year of each assessment; one row per named scenario; cryptocurrency removed where an assessment included it; emissions-only results converted to electricity (P01 section 2.4, p. 10)	base_year and scenario are recorded as stated; crypto_scope records the stated or inferred boundary and nothing is removed; no conversion between quantities is made.	recorded without harmonisation
Ledger fields with no P01 catalogue parameter	P01's metric is annual energy consumption in TWh (P01 section 1.2, p. 1)	unit and metric distinguish energy, capacity, capacity demand and growth; value_basis records table, text or chart values; shape records points, ranges and bounds; superseded_by and comparison_family record same-publisher vintages; SUPPORT_REVIEW.json binds an independent receipt to each row.	no P01 counterpart

Spread within like-for-like groups

Like-for-like group	Like-for-like: N (publishers); stated bounds	Same geography, any vintage or base year	Same metric, unit and year, any geography	Excluded, short of a supported receipt
Australia (National Electricity Market); consumption; TWh; 2030	1 (1); 15.5	1 (1); 15.5	15 (10); 6 to 1,065; open lower bound 1,200 (O27); adds O01, O02, O03, O05, O07, O08, O16, O17, O18, O27, C04, C05, C06, C07	none
Australia (National Electricity Market); consumption; TWh; 2036	1 (1); 33.7	1 (1); 33.7	1 (1); 33.7	none
China (IEA region, including Hong Kong); consumption; TWh; 2030	1 (1); 277	1 (1); 277	15 (10); 6 to 1,065; open lower bound 1,200 (O27); adds O01, O02, O03, O05, O07, O08, O16, O17, O18, O27, C02, C04, C06, C07	none
Denmark; consumption; TWh; 2030	1 (1); 6	1 (1); 6	15 (10); 6 to 1,065; open lower bound 1,200 (O27); adds O01, O02, O03, O05, O07, O08, O16, O17, O18, O27, C02, C04, C05, C06	none
Europe (IEA region); consumption; TWh; 2030	1 (1); 114	1 (1); 114	15 (10); 6 to 1,065; open lower bound 1,200 (O27); adds O01, O02, O03, O05, O07, O08, O16, O17, O18, O27, C02, C05, C06, C07	none
France; consumption; TWh; 2030	1 (1); 10 to 15	1 (1); 10 to 15	15 (10); 6 to 1,065; open lower bound 1,200 (O27); adds O01, O02, O03, O05, O07, O08, O16, O17, O18, O27, C02, C04, C05, C07	none
Global; consumption; TWh; 2030	3 (3); 945 to 1,065; open lower bound 1,200 (O27)	4 (3); 945 to 1,065; open lower bound 1,200 (O27); adds O01	15 (10); 6 to 1,065; open lower bound 1,200 (O27); adds O02, O05, O07, O08, O16, O18, C02, C04, C05, C06, C07	none
Global; capacity; GW; 2035	1 (1); 389	1 (1); 389	1 (1); 389	none
Global; capacity demand; GW; 2030	4 (3); 220 to 490.7	4 (3); 220 to 490.7	5 (3); 121 to 490.7; adds O22	none

Like-for-like group	Like-for-like: N (publishers); stated bounds	Same geography, any vintage or base year	Same metric, unit and year, any geography	Excluded, short of a supported receipt
Global; power demand growth; per cent growth; 2030; base year 2025	1 (1); 170	3 (1); 117 to 170; adds O10, O20	3 (1); 117 to 170	O11, O19
US; consumption; TWh; 2028	1 (1); 464	1 (1); 464	1 (1); 464	none
US; consumption; TWh; 2030	5 (5); 380 to 843	6 (5); 380 to 843; adds O02	15 (10); 6 to 1,065; open lower bound 1,200 (O27); adds O01, O03, O17, O27, C02, C04, C05, C06, C07	none
US; capacity; GW; 2030	1 (1); 151.734	1 (1); 151.734	1 (1); 151.734	none
US; capacity demand; GW; 2030	1 (1); 121	1 (1); 121	5 (3); 121 to 490.7; adds O09, O21, O24, C01	none
US; capacity demand; GW; 2035	1 (1); 176	1 (1); 176	1 (1); 176	none
US; installed capacity; GW; 2030	1 (1); 118	1 (1); 118	1 (1); 118	none
US; installed capacity; GW; 2035	1 (1); 194	1 (1); 194	1 (1); 194	O13

N counts observations, with the number of publishers in brackets; stated bounds are in the group's unit. Only 3 of the 17 like-for-like groups hold more than one observation. The Global consumption group for 2030 holds 3 observations from 3 publishers. The Global capacity demand group for 2030 holds 4 observations from 3 publishers. The US consumption group for 2030 holds 5 observations from 5 publishers and is not fully like for like, because it mixes cryptocurrency scopes (excluded, excluded by inference, included and not stated).

In each of the 3 groups that hold more than one observation, the like-for-like span is narrower than the span across all geographies for the same metric and year and equal to the span within the same geography, so the narrowing comes from separating geographies; adding other editions or base years within a geography leaves these spans unchanged. The narrowing therefore reflects the removal of figures that measure something else, not agreement among comparable forecasts.

The 4E review reports that studies it rated low (7 publications) gave global estimates for 2023 of 480 to 2,000 TWh across 17 scenarios and cases, and that studies it rated low-medium or higher (22 publications) gave 190 to 560 TWh across 43 scenarios and cases. Its executive summary gives 42 for the second count, and the per-class labels of its Figure ES.1 sum to 43. That result answers a different question: the 4E review filtered estimates of global energy use in 2023 by its own quality

rating, whereas this ledger groups 2025 and 2026 forecasts for later target years by stated quantity and boundary and rates no forecast's quality. With so few observations per group, and with one multi-member group mixing cryptocurrency scopes, whether like-for-like forecasts converge is not estimable from this sample.

Comparison limits

Annual electricity is energy used over a year, stated here in terawatt-hours; installed capacity, capacity demand and IT demand are power, stated in gigawatts; growth relative to a base year is a percentage whose meaning depends on the base. Converting one of these into another would need utilisation or baseline assumptions that the sources do not state, and the ledger supplies none.

Spans still combine scenario bounds with point estimates and differing cryptocurrency and facility boundaries. Their limits are not a probability interval, consensus estimate or validated like-for-like spread. Cryptocurrency appears among those boundaries because publishers draw the data-centre boundary differently: some publishers' data-centre figures include cryptocurrency mining, others exclude it and others leave it unstated, so a spread across them can mix boundaries. The ledger therefore records cryptocurrency coverage as one of the distinctions it retains rather than normalising it away.

Limits

No metered US national data-centre outcome series was collected. The package cannot provide forecast-accuracy scores, a calibrated correction factor or a capacity-uplift instrument. It also does not prove that public access is complete or that a source is current indefinitely.

The historical discussion of misquotations, earlier demand concerns and utility forecast errors requires its own source review before renewed use. Those topics do not supply a measured error distribution for this sample. Historical overprediction elsewhere cannot establish that a current data-centre forecast is too high. The earlier paper remains preserved in its published record; this version does not recertify its remaining narrative claims.

The comparison with the 4E review relies on the review's own transcription of the publications it catalogued; those publications were not retrieved for this version, and the review's counts of scenarios and cases differ between its executive summary and its conclusions. The candidate receipts rest on TrendForce's press releases and article and on Oxford Economics' summary page and AEMO's explainer of their joint forecast; TrendForce's underlying report and the joint forecast's full report were not retained. The non-US candidates rest on the IEA's 2026 report and 2025 data annex, the consumption chapter of RTE's Bilan prévisionnel 2025-2035 and the Danish Energy Agency's background note on data centres; RTE's 2030 values are read from a chart by the colour of their labels. The search for non-US figures on 11 October 2026 named the European Commission, ENTSO-E, RTE, EirGrid and Energinet; it did not cover other national operators, including those of Germany, the Netherlands, the other Nordic countries and Great Britain, or regions outside Europe beyond China, and an S&P Global page refused automated retrieval. The bounded search for publications dated 8 or 9 October 2026 cannot exclude a publication that search indexes had not yet

recorded, and values listed as known unregistered, including TrendForce's US figure, are not registered.

The reproduction checks described in the Verification note do not establish source truth or scientific validity. No forecast-quality or model-performance baseline was run.

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The historical claim that an earlier script mapped open directly to reproducible is unverified in this bundle because that script is not included. That label did not establish an executed reconstruction of the forecasting model. The revision separates a readable publication, an inspected figure, a description of the method and an actual model-reproduction attempt.

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AI assistance, conflicts and limitations

The research design, method, sourcing decisions and analytical judgements are the author's. OpenAI GPT-6 and Anthropic Opus 5.5 and Fable 5.1 assisted with data retrieval, calculation, literature search and drafting, so this text was artificially generated and was reviewed by the author before publication. The assisting models are not always neutral parties to the subject matter: OpenAI, whose GPT-6 model also reviewed the claim support for the candidate observations, Anthropic, and Google, whose Gemini 3.8 Flash High model reviewed the claim support for the registered observations and the hand-written sentences of this paper, all take part in the AI industry whose data-centre electricity demand these forecasts describe.

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What the author cannot guarantee: a language model's output can be wrong in ways that survive review. In prose the error is locally plausible and consistent in tone with what surrounds it; in code it simply runs, and a wrong constant or a mis-set filter still returns a clean number. Several methods have been deployed to mitigate this, including explicit instructions, internal red-teaming and cross-lab blindspot checks, but the author does not claim the review is exhaustive. Corrections are logged against the DOI when surfaced.

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